Types of sampling methods

Learning Objectives

By the end of this session, you will be able to:

- Understand the concept of sampling.
- Understand different types of sampling and their use case.
- Implementation of sampling techniques in Python!

What is Sampling?

- Selecting a subset from a larger population.
- Saves time, cost
- Enables conclusions.

Why is Sampling Important?

- ✅ Populations too large to fully study
- ✅ Reduces effort
- ✅ Provides valid results

Random Sampling

- Each member has equal chance.

- Simple Random Sampling is used when you want a straightforward and fair representation of the entire population.

$$P(\text{select an item}) = \frac{n}{N}$$

- Randomly picking 10 passengers from 100 Titanic passengers. Each has a 10% chance of being selected.

Stratified Sampling

- In **stratified sampling**, the population is
  - ✓ **divided into groups (strata)**
  - ✓ based on an important characteristic (like gender), and
  - ✓ then **samples are drawn proportionally from each group**.
- This ensures your sample **respects the structure** of the population.
- Formula: (stratum size / population size) × sample size.

Stratified Sampling

- A college has 800 students: 600 UG, 200 PG.

You want to sample 100 students using stratified sampling technique. How many students from UG and PG will be in your sample?

Stratified Sampling

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Undergrad sample =

$$\frac{600}{800} \times 100 = 75$$

Postgrad sample =

$$\frac{200}{800} \times 100 = 25$$

Systematic Sampling

- In **systematic sampling**, you **select every k-th item** from an ordered list.
- It’s simple, fast, and effective when your data is organized.
- Select every k-th element.
- Formula: $k = N / n$
- Example: Survey every 10th customer entering a store.

Cluster Sampling

- Divide into clusters.
- Randomly select whole clusters.
- Example: Randomly selecting 2 out of 5 schools and surveying **all students** inside.

Comparison

Sampling Method	Key Idea	Best Use Case
Random Sampling	Equal chance for all	General surveys, experiments
Stratified Sampling	Group by characteristic, then sample	When important subgroups exist
Systematic Sampling	Every k-th item selected	Assembly lines, orderly populations
Cluster Sampling	Randomly pick entire groups	When groups are naturally formed

Python Implementation

MCQ

1. Which sampling ensures proportional representation?

A) Random

B) Stratified

C) Cluster

2. Which method selects every k-th item?

A) Random

B) Systematic

C) Cluster

MCQ

1. Which sampling ensures proportional representation?

A) Random

B) Stratified ✅

C) Cluster

2. Which method selects every k-th item?

A) Random

B) Systematic ✅

C) Cluster

Session Summary

In this session, we have discussed about:

- Understand the concept of sampling.
- Understand different types of sampling and their use case.
- Implementation of sampling techniques in Python!

Reference Links

- Complete Guide to Sampling Methods: Random, Stratified, Systematic, and Cluster Sampling with Python Examples
- Analyzing loan data with binomial and poisson distribution
- 10 Must-Know Probability Concepts for Every Aspiring Statistician and Data Scientist
- Top 20 Interview Questions on Descriptive and Inferential Statistics
- 10 Must-Know Statistics Interview Questions (With Real Examples & Formulas!)